

BIO

Respiration in Organisms

Respiration: It is the process in which food is oxidised in the cells to release energy.

It occurs in two steps-

a) External Respiration or Breathing -

• It is the process of exchange of respiratory gases between the body of the living organism and its surroundings (Physical Process).

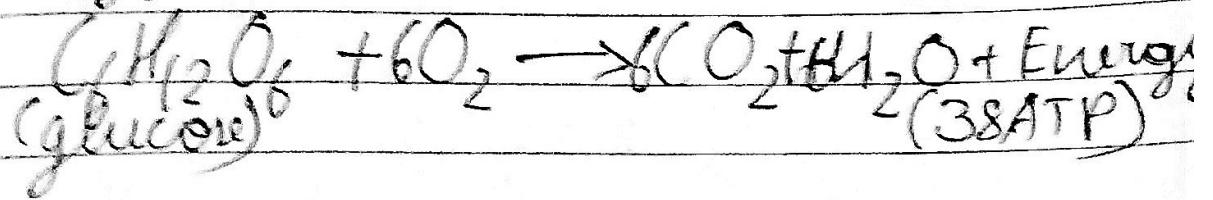
b) Internal or Cellular Respiration -

• It is a biochemical process. In this process food (glucose) is oxidised in the cells to release energy.

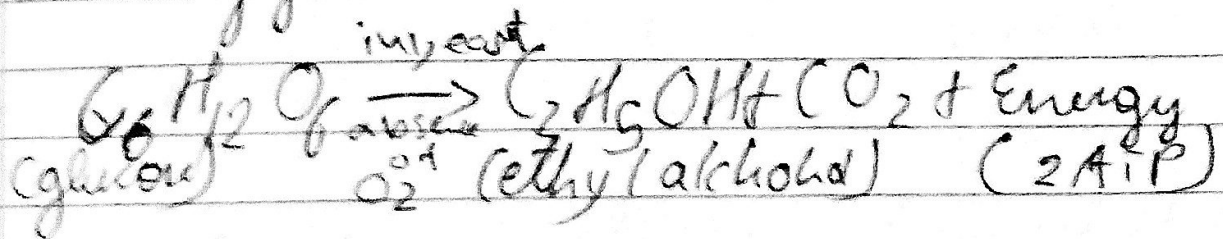
The energy released during respiration is stored in the form of ATP (Adenosine triphosphate)

Types of cellular respiration -

- (i) Aerobic respiration - In this process food is oxidised in the presence of Oxygen to release energy.



- (ii) Anaerobic respiration - In this process food is oxidised in the absence of Oxygen to release energy.



During a rigorous physical exercise, the muscle cells contract and relax rapidly for which they require more amount of energy. The muscle cells then begin to respire anaerobically also. During this anaerobic respiration, lactic acid is produced.

along with energy.

Glucose $\xrightarrow[\text{absence of } O_2]{\text{in muscle cells}}$ Lactic acid + Energy (2 ATP)

- The accumulation of lactic acid in the muscle cells may result in muscle cramps.